

A Continuous Perturbation Without a Two-Sided Functional Alexandrov Derivative

Candidate counterexample for arXiv:2502.16929

July 23, 2026

Status. Candidate full counterexample, likely valid pending expert review. The target is the unrestricted continuous-perturbation extension discussed in Section 2 of the source, not the paper’s solved functional Minkowski theorem.

1 Source question

Falah and Rotem study, for a convex function ϕ and a perturbation ξ , the derivative

$$\left. \frac{d}{dt} \right|_{t=0} \int_{\mathbb{R}^n} e^{-(\phi^* + t\xi)^*}. \quad (1)$$

They prove it for bounded continuous ξ and for support functions. Between those results they state that computing (1) “for all continuous functions $\xi : \mathbb{R}^n \rightarrow \mathbb{R}$ ” is a nontrivial task which they do not carry out [1]. The full passage appears on PDF page 9.

2. A PARTIAL ALEXANDROV LEMMA

In our solution of the Minkowski problem we will need to compute derivatives of the form

$$(2.1) \quad \left. \frac{d}{dt} \right|_{t=0} \int_{\mathbb{R}^n} e^{-(\phi^* + t\xi)^*},$$

where $\phi \in \text{Cvx}_n$ and $\xi : \mathbb{R}^n \rightarrow \mathbb{R}$ is a “nice enough” function. The function $\phi^* + t\xi$ is not necessarily convex, but we still compute its Legendre transform using the standard formula, i.e.

$$(\phi^* + t\xi)^*(y) = \sup_{x \in \mathbb{R}^n} (\langle x, y \rangle - (\phi^*(x) + t\xi(x))).$$

Now that if $\xi \in \text{Cvx}_n$ and if we define $f = e^{-\phi}$ and $g = e^{-\xi^*}$, then $e^{-(\phi^* + t\xi)^*} = f \star (t \cdot g)$ for all $t > 0$. Therefore in this case the derivative we are trying to compute is exactly the one from the first variation formula (1.1). However, formula (1.1) concerns only positive values of t (i.e. the right derivative), and for us it would be crucial to have a two-sided derivative.

In the case that ϕ^* is the support function of a convex body and ξ is continuous and 1-homogeneous, the computation of (2.1) is exactly the well-known Alexandrov lemma, which is a standard ingredient in the proof of the classical Minkowski problem. Our goal is therefore to prove a functional extension of this lemma. As was already mentioned in the introduction such a functional Alexandrov lemma was recently proved in [34], but its assumptions are not suitable for our goals.

Computing the derivative (2.1) for *all* continuous functions $\xi : \mathbb{R}^n \rightarrow \mathbb{R}$ is a non-trivial task, which we do not need for our goal so we will not carry out here. Instead we will only discuss two partial cases. The first case is relatively straightforward and was already computed in [29 Proposition 5.4]:

The example below shows that continuity alone cannot guarantee existence of the two-sided derivative.

2 Definitions

For $q : \mathbb{R} \rightarrow (-\infty, +\infty]$, its Legendre transform is

$$q^*(x) = \sup_{y \in \mathbb{R}} \{xy - q(y)\}.$$

Write $\mathbf{1}_A^\infty$ for the convex-analysis indicator which is zero on A and $+\infty$ outside A .

3 Proof intuition

Start with the indicator of an interval. Its dual potential is $|y|$, whose slope at infinity determines the domain of its conjugate. Add a continuous linear-growth perturbation whose ratio to $|y|$ oscillates forever between -1 and 1 . For either sign of t , the smallest asymptotic slope becomes $1 - |t|$ and is attained along an unbounded sequence. The conjugate is therefore the indicator of an interval whose radius is $1 - |t|$. Its integral shrinks linearly in $|t|$, producing a cusp at zero.

4 Counterexample

Theorem 1. *There exist $\phi \in \text{Cvx}_1$ and a continuous function $\xi : \mathbb{R} \rightarrow \mathbb{R}$ with $|\xi(y)| \leq |y|$ such that $\phi^* + t\xi$ is coercive for every $|t| < 1$, the integral*

$$I(t) = \int_{\mathbb{R}} e^{-(\phi^* + t\xi)^*(x)} dx$$

is finite for every $|t| < 1$, but I is not differentiable at zero. More precisely, $I(t) = 2(1 - |t|)$.

Proof. Set

$$\phi = \mathbf{1}_{[-1,1]}^\infty, \quad \xi(y) = |y| \sin(\log(1 + |y|)).$$

The function ϕ is convex and lower semicontinuous, and $\int e^{-\phi} = 2$, so $\phi \in \text{Cvx}_1$. Directly from the definition,

$$\phi^*(y) = \sup_{x \in [-1,1]} xy = |y|.$$

The function ξ is continuous and satisfies $|\xi(y)| \leq |y|$.

Fix $|t| < 1$ and write

$$q_t(y) = \phi^*(y) + t\xi(y) = |y| [1 + t \sin(\log(1 + |y|))], \quad a_t = 1 - |t| > 0.$$

For every y ,

$$q_t(y) \geq a_t |y|,$$

so q_t is coercive. If $|x| \leq a_t$, then

$$xy - q_t(y) \leq |x| |y| - a_t |y| \leq 0.$$

Equality holds at $y = 0$, hence $q_t^*(x) = 0$.

It remains to consider $|x| > a_t$. If $t > 0$, set

$$r_k = \exp(3\pi/2 + 2\pi k) - 1;$$

then $\sin(\log(1 + r_k)) = -1$ and $q_t(r_k) = a_t r_k$. If $t < 0$, instead set

$$r_k = \exp(\pi/2 + 2\pi k) - 1;$$

then $\sin(\log(1 + r_k)) = 1$ and again $q_t(r_k) = a_t r_k$. For $x > a_t$, in either case

$$x r_k - q_t(r_k) = r_k(x - a_t) \longrightarrow +\infty.$$

For $x < -a_t$, use $y = -r_k$ and the evenness of q_t to obtain the same conclusion. Therefore

$$q_t^*(x) = \begin{cases} 0, & |x| \leq a_t, \\ +\infty, & |x| > a_t. \end{cases}$$

Consequently,

$$I(t) = \int_{\mathbb{R}} e^{-q_t^*(x)} dx = |[-a_t, a_t]| = 2(1 - |t|).$$

Thus $I'_+(0) = -2$ whereas $I'_-(0) = 2$, and the two-sided derivative does not exist. $\square$

5 Verification, scope, and novelty

Every object in the example meets the literal hypotheses: ϕ lies in the source class, ξ is continuous and real-valued, and the perturbed dual potential is coercive for both signs of every sufficiently small t . The failure is caused only by the absence of a limiting recession slope for ξ .

This does not affect Proposition 2.1 of the source, where ξ is bounded, or Theorem 2.3, where ξ is a support function. It also leaves open general results under cosmic continuity or other recession regularity.

The run's lightweight indexes and bounded web searches on July 23, 2026 were checked using arXiv:2502.16929, the exact title and authors, “functional Alexandrov lemma,” “all continuous functions,” “two-sided derivative,” and close Legendre-perturbation variants. No later answer or matching counterexample was found. Novelty confidence is moderate pending specialist review.

6 Human review recommendation

The main check is the exact conjugate-domain argument, especially the two unbounded sequences on which the oscillating coefficient attains $1 - |t|$. The reviewer should also confirm that the source phrase “all continuous functions” is intended literally, rather than as shorthand for an unstated class with a recession limit.

References

- [1] T. Falah and L. Rotem, *On the functional Minkowski problem*, arXiv:2502.16929 (2025).